

Harikesh Byrandurga Gopinath

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Education

- The Ohio State University, Columbus, OH, USA** **May 2026**
Master of Science in Computer Science and Engineering [GPA: 3.71 / 4.0]
- Siddaganga Institute of Technology, India** **Aug 2021**
Bachelor of Engineering in Information Science and Engineering [GPA: 8.74 / 10.0]

Technical Skills

Programming Languages: Java, Python.

Frameworks: Spring Boot, JUnit, Mockito, Flask.

Databases : PostgreSQL, MySQL, Firebase, MongoDB, SQLite.

Others: Jenkins, SonarQube, Git, Postman, RabbitMQ,

Android Studio, Swagger.

Professional Experience

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| Graduate Research Associate | The Ohio State University | Aug 2025 - Present |
| <ul style="list-style-type: none">Building an end-to-end software pipeline that automates data collection, transfer from edge devices on agricultural sites, and coordinates HPC jobs to launch the appropriate processes on the incoming data automatically.Globus, Tapis, Python, Flask, React, PostgreSQL. | | |
| Software Engineer II | Deloitte | July 2022- July 2024 |
| <ul style="list-style-type: none">Built and optimized a Java/Spring Boot + PostgreSQL account-search backend for 100M+ records, using trigram indexes, partitioning, and DevOps-aligned DB tuning to cut query times from 4–5 minutes to under 1 second.Integrated SonarQube and Jenkins to build a local CI pipeline for an internal project, and incorporated the ChatGPT API to enhance the application's output. | | |
| Software Engineer I | Deloitte | July 2021- July 2022 |
| <ul style="list-style-type: none">Refactored a Java microservice for a health-insurance platform to delta-sync JSON to MySQL (change detection + batch upserts), streamlining the flow and reducing execution overhead ~20% on ~100k-record runs.Built REST APIs for a critical records-sync module, adding out-of-sequence insurance plan version handling (edge-case detection, idempotent updates) to ensure accurate synchronization across scenarios; partnered with the client on POCs and delivered the feature end-to-end from scratch. | | |
| Software Engineer Intern | Deloitte | March 2021 - July 2021 |
| <ul style="list-style-type: none">Trained in Java, Spring Boot, Angular, and MuleSoft; built Spring Boot microservices to analyze mutual-fund portfolios, generate drift/overlap reports with user notifications, and preprocessed Excel inputs using Python/Pandas—project ranked top 5/30 at the expo and was showcased to company executives. | | |

Achievements

- Received spot award for handling critical tasks in a client project, HashedIn By Deloitte, April, 2022.

Research Projects

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| AI for Distributed Protocols | Jan 2024 - May 2024 |
| <ul style="list-style-type: none">A machine learning model that predicts the actions to be taken by the nodes in a distributed system.Parallelized the model verifier using Z3 and OpenMP, achieving a 2× speed-up in generating examples. [OpenMP, Z3, C++] | |

Publications

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| Neural machine translation for regional languages | Jan 2021 - Jul 2021 |
| Tools and Technologies: Python, Keras, Pandas | |
| <ul style="list-style-type: none">Built a Neural Machine Translation system for chat apps enabling Hindi↔English communication; implemented Keras-based translation APIs with pandas-driven CSV preprocessing; published “Neural Technique for Language Translation,” IJISAE 10(4), 2022, ISSN 2147-6799, DOI: 16-12-2022. | |